POWER PULSE

Equal Area Criterion – Cases Important Formulae

Notes by BALAJI SIR

1 Angles

1.
$$\delta_0 = \sin^{-1}\frac{P_m}{P_{max1}}, \qquad \delta_{max} = \pi - \sin^{-1}\frac{P_m}{P_{max3}}$$

P_max1 pre-fault, P_max2 during fault, P_max3 post-fault

2 Areas

2.
$$A_1 = \int_{\delta_0}^{\delta_c} (P_m - P_{max2}\sin\delta)\,d\delta = P_m(\delta_c - \delta_0) + P_{max2}(\cos\delta_c - \cos\delta_0)$$

3.
$$A_2 = \int_{\delta_c}^{\delta_{max}} (P_{max3}\sin\delta - P_m)\,d\delta, \qquad \text{stable if } A_2 \geq A_1$$

3 Critical clearing angle

4.
$$\cos\delta_c = \frac{P_m(\delta_{max} - \delta_0) + P_{max3}\cos\delta_{max} - P_{max2}\cos\delta_0}{P_{max3} - P_{max2}}$$

5.
$$P_{max2} = 0: \quad \cos\delta_c = \frac{P_m}{P_{max3}}(\delta_{max} - \delta_0) + \cos\delta_{max}$$

terminal 3-φ fault

6. Same pre/post fault, $P_m = \frac{1}{2}P_{max}$: $\delta_0 = 30^\circ,\ \delta_{max} = 150^\circ,\ \delta_c = 79.6^\circ$

4 Critical clearing time

7.
$$t_c = \sqrt{\frac{2H(\delta_c - \delta_0)}{\pi f\, P_m}} \qquad (P_e = 0 \text{ during the fault; angles in rad})$$

depends on H; the angle δ_c does not

5 Sudden changes

8. Sudden load step $P_0 \to P_1$: $P_1(\delta_m - \delta_0) = P_{max}(\cos\delta_0 - \cos\delta_m)$

9.
$$\text{Limit: } \delta_m = \pi - \sin^{-1}\frac{P_1}{P_{max}} \quad \text{(sudden step or line outage)}$$